

Call Manager (Aegis Calling) — Current Features

Document type: Feature catalog (as-built, grounded directly in the codebase) **Document version:** 1.0 **As of:** July 2026, deployment1 branch

1. Overview

Call Manager is a live, working AI outbound-calling platform for The Shri Ram Academy (TSRA) admissions outreach. This document describes what the system **actually does today** — every item below is implemented and running, not planned or aspirational.

2. AI Voice Agent ("Maya")

- **Single shared voice agent** across every campaign, driven entirely by one prompt file (agent_prompt.md) — no per-campaign forks or duplicated logic.
- **Voice:** ElevenLabs 11labs-Monika (Indian-accented female voice).
- **Multilingual conversation:** understands and replies in **English, Hindi, and Tamil**, including natural code-switching ("Hinglish"). If a caller speaks Telugu, the agent plainly says it cannot understand and offers to continue in Hindi or English rather than guessing.
- **Identity check opening:** every call opens with a fixed "Hi, am I speaking with {name}?" and recognizes a wide range of colloquial/regional affirmatives ("ha", "haan", "achha", "aama," etc.), including short or garbled variants.
- **Grounded answers only:** every factual claim about the school is backed by a real-time knowledge lookup (native Retell knowledge base + a custom lookup_school_info tool) — never invented from memory.
- **One question at a time:** during booking, the agent asks for date/time, then purpose, then email — separately, waiting for a real answer each time — rather than bundling multiple questions into one confusing turn.
- **No fabricated confirmations:** a spoken "your appointment is booked" or "callback scheduled" is only ever said after the corresponding backend tool call has actually run and returned success.
- **Natural, human tone:** varied phrasing rather than repetitive scripted politeness ("could you kindly..." on every single line was deliberately removed in favor of natural variation).

Available tools the agent can call mid-conversation

Tool	Purpose
lookup_school_info	Real-time factual lookup from the scraped school website
schedule_callback	Creates a callback for a specific resolved time
book_appointment	Books a campus visit / admission counseling slot
mark_outcome	Records the final call outcome
end_call	Hangs up

3. Outbound Calling

- **Campaign-based dialing:** admin uploads an Excel/CSV lead list, which becomes a campaign ("UploadBatch"); the system dials contacts up to a configurable concurrency limit.
 - **Independent per-campaign queues:** one campaign's pace never blocks another's.
 - **Auto-retry with a cap:** if a call is unanswered or ends unresolved, the system automatically retries — capped at 2 automatic retries (3 total attempts) — before flagging the contact for manual follow-up.
 - **Do-Not-Call enforcement:** contacts marked DoNotCall are excluded from all future dials, campaign or manual.
 - **Manual "Call Now":** staff can trigger an immediate call to any single contact from the dashboard.
 - **Busy-contact protection:** a contact already on an active call cannot be dialed again by a different trigger (another scheduled callback, a manual call, or a campaign dial) — the second attempt defers instead of double-ringing the same person.
-

4. Appointment Booking

- Collects and confirms date, time, purpose, and email before booking.
 - **Idempotent booking:** re-booking for a contact with an existing "Booked" appointment updates that same record instead of creating a duplicate.
 - **Zero added call latency:** Google Calendar sync and confirmation email are dispatched as background tasks — the live call is never held up waiting on either.
 - **Email correction handling:** if the caller says the on-file email is wrong, the agent must obtain and confirm a new one before booking — it will not book with an empty or unconfirmed email.
-

5. Reschedule / Callback Flow

- Resolves relative time expressions ("in 10 minutes," "tomorrow evening") against the actual call timestamp, in IST.
 - Confirms the resolved time back to the caller before scheduling.
 - A callback is only ever confirmed to the caller after the scheduling tool call has actually succeeded.
 - Fires automatically at the scheduled time via a background scheduler (APScheduler), independent of the live dashboard being open.
-

6. Knowledge Base

- The school's website is scraped into searchable content chunks.
- **Automatic nightly refresh** (3 AM IST) plus an on-demand "Refresh Now" button in the dashboard.
- **Native Retell knowledge base** is attached to the agent for fast, semantically-aware retrieval, with the custom keyword-search tool as a fallback.

- Dashboard includes a **test-search** box so staff can directly query the knowledge base and see what the agent would find for a given question.
-

7. Dashboard (React frontend)

Dashboard (overview) page

- At-a-glance view combining contacts, recent call history, upcoming appointments, and pending callbacks in one place.

Campaigns page

- Lists all uploaded campaigns.
- Expand any campaign to see its per-contact call history.
- Trigger an individual call directly from the campaign view.
- View full transcripts inline.

Contacts / Leads page

- Search and filter contacts by status.
- Per-contact call history, transcript, and recording playback.
- Manual reschedule action per contact (set a new callback time directly).

Scheduling page

- Two views in one page: **Callbacks** and **Appointments**, switchable via tabs.
- Edit/reschedule an existing callback's time.
- Manually create a new scheduled callback.

Settings page (admin only)

- Configure Retell API key, phone number, agent ID.
- Configure Google Calendar service-account credentials and calendar ID.
- Configure SMTP host/port/username/password/from-address.
- Configure dialer concurrency limit, max retry attempts, retry backoff hours.
- Configure the tool-webhook shared secret (`aegis_tools_secret`).
- Secrets are masked in the UI once saved.

Login page

- Email/password authentication issuing a JWT session token.
- Role separation: admin (full access) vs staff (no Settings access).

8. Backend Reliability Features

- **Self-configuring voice agent:** on every backend startup, the shared Retell agent's prompt, tools, and webhook URL are automatically synchronized to point at *that* deployment — no manual setup script needs to be run on a fresh deployment, ever. The agent/LLM identity is persisted in the database (not a local file), so this works identically whether deployed on Coolify, AWS, or anywhere else.
 - **Resilient database connections:** TCP keepalives, pre-flight connection validation, and periodic connection recycling — protects against a remote database silently dropping idle connections and causing tool-call timeouts.
 - **No silent failure masking:** a failed call-placement request now surfaces the real error (e.g. a genuine Retell API rejection) instead of being hidden behind a fabricated "success" response — and a contact's status correctly reverts if the call was never actually placed, instead of getting stuck.
 - **Webhook signature verification:** incoming Retell/Cal.com webhooks are HMAC-verified when a real signing secret is configured.
 - **Background job scheduler** running four recurring jobs: safety status reset, callback dialer sweep (every minute), nightly knowledge refresh, and Google Calendar reconciliation (every 10 minutes).
-

9. Integrations

System	What it's used for
Retell AI	Voice calls, speech-to-text/text-to-speech, LLM-driven conversation
Google Calendar API	Creates a calendar event for every booked appointment (via service account)
Gmail SMTP	Sends styled HTML confirmation emails for bookings
TSRA website	Source content for the knowledge base (scraped, not hand-maintained)

10. Data Tracked (7 core tables)

Table	What it stores
contacts	Leads — name, phone, email, status
upload_batches	Campaigns
call_attempts	Every call placed — outcome, duration, transcript, recording
scheduled_callbacks	Pending and fired reschedule requests
appointments	Booked campus visits with Google Calendar links
knowledge_chunks	Scraped website content

Table	What it stores
settings	Runtime credentials/config, including the agent's persisted identity

11. Security Features Currently Implemented

- JWT-based session authentication with admin/staff role separation.
- Secrets masked in the Settings UI once saved.
- Optional shared-secret header (`aegis_tools_secret`) to authenticate incoming tool-webhook calls.
- Webhook signature verification when a real signing key is configured.

This document describes the system as it exists in the codebase today — every feature listed has been implemented and exercised through live testing, not merely designed.